

- 4 A nucleus P undergoes α -decay to form nucleus Q.

(a) Complete the equation for this decay.



(b) (i) State the principle of conservation of momentum.

.....

 [2]

(ii) Before the decay, nucleus P has a speed of $3.2 \times 10^5 \text{ m s}^{-1}$. After the decay, nucleus Q is stationary.

Calculate the speed of the alpha particle after the decay.

speed = m s^{-1} [2]

[Total: 6]